The research loop

Staged experiments, judged by a rule fixed in advance; code computes when to stop.

Limitation census
flaws of the reference method

Two sealed idea streams
committed, then revealed

Preregistration
evaluators, comparison rule, stop rule

STAGED EXPERIMENTS

next generation

Baseline
the reference

Screen
promote or discard

Full
rounds of candidates

Select
commit k of N trials

Confirm
once, fresh seeds

adaptive screen

withheld, sealed until selection

adaptive full

Ablation
one component

Objection test
one concern

confirmation never shares population, seeds or generation with selection

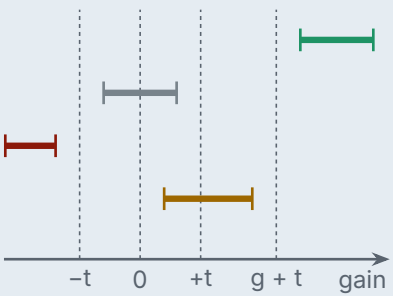
DECISION PER CELL: GAIN INTERVAL [a, b]

better a > g + t

equivalent |a|, |b| ≤ t

worse b < −t

unknown otherwise



CODE STOPS THE LOOP

1 successes and coverage met

2 round limit, all observed

3 budget spent

coverage unmet: reported as a null funnel

Critique
one experiment per concern

Acceptance
gains attributed by ablation

Meta-review
every claim and failure

Integrity
fresh re-execution

objections return to experiments

Key: Agents · RAVEL · Data · Checks and review · Rule